Kori
Pickle

← BACK TO PROJECTS

PORTFOLIO PROJECT 17 · 2026-2027

Integrated Care Journey & Responsible AI Control Intelligence™

A student-developed healthcare operations case study that converts a broad future-healthcare vision into a practical control system for interoperability, remote access, automation, responsible AI, cybersecurity, patient communication, exception ownership, synthetic UAT, and verified closed-loop completion.

Created by Kori Pickle, BSHA Candidate, University of Phoenix. Educational, simulated, and no PHI. No real patient, employer, payer, claims, EHR, portal, device, or operational outcome data were used.

REVIEW SIMULATED CASE

DOWNLOAD BRANDED PDF

HEALTHCARE OPERATIONS INTELLIGENCE ENGINE™

Interoperability
Remote Access
Automation
Responsible AI

CORE QUESTION

Where did the workflow first lose control?

Kori Pickle · BSHA Candidate

EXECUTIVE BREAKDOWN

The image describes a technology vision. Operations determines whether it works.

Integrated systems

Connected applications are not enough. Information must retain meaning, source, identity, access rules, and an actionable destination.

Closed-loop execution

A message is not a completed handoff. The receiving owner must accept, act, communicate, escalate exceptions, and verify closure.

Responsible AI

AI may support approved administrative tasks, but it requires source control, human review, traceability, uncertainty handling, and a fallback.

Remote and human-centered access

Virtual access must address technology, language, disability, privacy, clinical suitability, patient preference, and an alternative pathway.

> Integration is not the number of connected tools. It is the reliability of the data, handoffs, ownership, safeguards, patient access, and verified completion across the journey.

FUTURE-STATE CAPABILITY MAP

Seven capabilities - and the hidden controls behind each one.

Interoperability

Identity matching, usable exchange, provenance, semantic consistency, receipt confirmation, and workflow placement.

Remote access

Identity, consent, accessibility, technical support, privacy, suitability, and an in-person fallback.

AI assistants

Approved use case, limited sources, draft status, human review, traceability, and escalation.

Automation

Visible rules, retries, exception queues, change control, monitoring, and a manual recovery path.

Patient journey coordination

One owner, accepted handoffs, clear deadlines, consistent instructions, and evidence of completion.

Cyber resilience

Access controls, backups, downtime procedures, incident response, recovery validation, and reconciliation.

> Measurement is the seventh capability: the system must prove whether access improved, handoffs closed, exceptions resolved, AI outputs were reviewed, and patients reached the next step.

SIMULATED NO-PHI CASE

The connected journey that still failed.

A simulated patient is discharged after a complex hospital stay. The organization uses an EHR, payer portal, telehealth platform, patient portal, pharmacy system, remote-monitoring vendor, and an AI-enabled administrative assistant. Each tool works individually, yet the patient misses a follow-up visit, the authorization task ages, a device alert is not acknowledged, and the patient receives conflicting instructions.

Missed follow-up	The discharge task was sent, but no receiving owner accepted responsibility.
Authorization delay	The payer requirement existed in a separate system and was not linked to scheduling readiness.
Unacknowledged alert	The remote-monitoring alert entered a vendor dashboard without a defined operational owner or escalation path.
Conflicting messages	Portal, text, and AI-generated reminders used different source data.
First loss of control	The journey moved from discharge to ambulatory follow-up without one shared status model, one accepted owner, and one source-controlled patient instruction.

JOURNEY ORCHESTRATION BLUEPRINT

Eight stages from identity to verified closure.

01	02	03	04	05	06	07	08
IDENTIFY	CONNECT	INTERPRET	ASSIGN	EXECUTE	COMMUNICATE	ESCALATE	CLOSE

> "Sent" is not "received." "Received" is not "accepted." "Accepted" is not "completed." The journey closes only when the next step is verified or an authorized exception is documented.

CONTROL REDESIGN

Integrated Care Journey Control Matrix™

Control domain	Required design	Status and evidence	Escalation trigger
Patient identity	Enterprise matching, duplicate review, demographic and contact reconciliation.	Confirmed, possible duplicate, conflict; source and timestamp.	Identifiers conflict or multiple records appear.
Data exchange	Standards-based exchange, provenance, receipt confirmation, semantic mapping.	Sent, received, interpreted, rejected; interface log.	Message delayed, incomplete, rejected, or mapped incorrectly.

Control domain	Required design	Status and evidence	Escalation trigger
Access and consent	Role-based access, approved purpose, minimum necessary use, permission where applicable.	Authorized, limited, expired, unresolved.	User, purpose, or data scope does not match approval.
Journey ownership	Single accountable owner, acceptance, due date, backup owner, escalation threshold.	Assigned, accepted, in progress, blocked, closed.	No owner, missed acceptance, aging, or conflict.
Responsible AI	Approved use, limited source, human review, traceability, monitoring, fallback.	Draft, reviewed, approved, rejected; reviewer and source.	Missing source, low confidence, conflict, bias, or patient impact.
Automation	Visible rules, retries, exception queue, change control, manual fallback.	Completed, failed, retry, exception, manual.	Silent failure, duplicate action, repeated error, unavailable dependency.
Remote access	Accessibility, language, device support, privacy, suitability, alternate channel.	Ready, support needed, in-person required, declined.	Patient cannot connect, lacks privacy, or needs in-person care.
Cyber resilience	Access control, backup, downtime procedures, incident escalation, recovery reconciliation.	Normal, degraded, downtime, recovered, reconciled.	Suspicious activity, outage, data loss, or unreconciled work.

RESPONSIBLE AI BOUNDARY

AI should support the workflow without becoming an invisible decision-maker.

Appropriate administrative support

Draft a non-clinical reminder, summarize an approved policy, suggest a queue category, identify missing fields, create a meeting recap, or surface an aging task - with human review.

High-risk or inappropriate use

Independently diagnose, determine medical necessity, approve coverage, make a final compliance judgment, send unreviewed patient instructions, or silently close a patient-impacting task.

AI control	Portfolio question
Purpose	What exact task is the tool permitted to support?
Source	Which approved data or document produced the output?
Human review	Who must review it before use, communication, or closure?
Uncertainty	How does the tool show missing data, confidence limits, or conflict?
Fairness and accessibility	Could the output disadvantage a population or create language or access barriers?
Monitoring and fallback	What errors, overrides, incidents, and unavailable-tool scenarios are reviewed?

> AI does not create workflow control. A controlled workflow determines when AI may help, who must review it, and what evidence is required before the work moves forward.

SYNTHETIC UAT

Test the exceptions before patients and staff absorb the defects.

Duplicate identity

Stop automated outreach, create an owned identity exception, and preserve the audit trail.

Outdated contact data

Reopen the task when outreach fails and do not count a sent message as successful communication.

Interface delay

Expose the missing receipt, age the exception, and launch an approved fallback route.

Authorization not linked

Keep readiness unresolved until evidence in the payer and scheduling workflows is reconciled.

AI summary conflict

Reject the output, record the reason, and prevent downstream use.

Remote-access barrier

Offer technical support, accessible formats, language help, or an alternate channel.

Unowned alert

Route the alert to a defined queue and escalate based on priority and age.

Cyber downtime

Use a controlled manual process and reconcile all work after recovery.

MEASUREMENT FRAMEWORK

KPIs that show whether integration is producing control - not just activity.

KPI	Definition	Why it matters
Closed-loop journey completion	Completed next steps divided by journeys due for completion.	Measures whether patients reached the intended operational milestone.
Orphan task rate	Open tasks without an accountable owner divided by all open tasks.	Reveals where connected systems create work without responsibility.
Handoff acceptance time	Time from routing to receiving-owner acceptance.	Separates sending from transfer of responsibility.
Interface exception aging	Median and oldest unresolved exchange exception.	Shows data that is connected but not operationally usable.
Patient contact success	Confirmed successful contacts divided by outreach attempts.	Prevents “sent” from being mistaken for communication.
Digital access resolution	Remote barriers resolved by due date divided by barriers identified.	Measures whether remote access is genuinely accessible.
AI verification rate	AI-supported outputs reviewed before use divided by outputs used.	Confirms that human review is part of the process.
Downtime reconciliation	Manual records reconciled after recovery divided by manual records created.	Protects continuity and data integrity after outages.

> All targets in this educational project are simulated design assumptions - not employer results, savings, or patient outcomes.

CAREER VALUE

Recruiter-visible proof for 2026-2027 roles.

Patient access and operations

Identity, communication needs, appointment readiness, access barriers, ownership, and escalation.

Health IT and implementation

Business rules, fields, interfaces, status design, defect testing, training, adoption, and audit trails.

Quality improvement

Root-cause mapping, first-loss-of-control analysis, KPIs, corrective action, and recurrence review.

Responsible AI support

Approved use boundaries, source evidence, review checkpoints, exception categories, and monitoring questions.

> "I translated a future-healthcare vision into a simulated operational control model covering interoperability, remote access, workflow ownership, responsible AI, automation exceptions, cybersecurity, synthetic UAT, and closed-loop KPIs - without claiming clinical, compliance, or system-administration authority."

HEALTHCARE OPERATIONS | INTEROPERABILITY | RESPONSIBLE AI | IMPLEMENTATION

PATIENT EXPERIENCE | CYBER RESILIENCE | SYNTHETIC UAT | QUALITY IMPROVEMENT

OFFICIAL GUIDANCE

Primary sources informing this educational project.

ONC / ASTP - TEFCA · Nationwide framework and common floor for secure health-information exchange.

ONC / ASTP - Interoperability · Health information exchange, patient access, and coordinated care.

CMS Interoperability and Prior Authorization Final Rule · Patient, provider, payer, and prior-authorization API requirements and operational changes.

ONC HTI-1 Final Rule · Interoperability, algorithm transparency, and information sharing.

NIST AI Risk Management Framework · Voluntary framework for managing AI risks.

HHS Healthcare and Public Health Cybersecurity Performance Goals · Healthcare-specific cybersecurity preparedness and resilience.

Telehealth.HHS.gov · Remote and hybrid care access, patient preparation, privacy, and technical support.

BRANDED CASE STUDY

Download the complete 12-page PDF.

The PDF includes the full breakdown, simulated failure case, eight-stage journey blueprint, control matrix, responsible-AI boundary, synthetic UAT plan, KPI framework, 30-day simulated improvement plan, recruiter language, resume bullet, and official sources.

DOWNLOAD PDF

RETURN TO PORTFOLIO

Created by Kori Pickle · BSHA Candidate · Student-developed · Simulated · No PHI

Healthcare Operations Intelligence Engine™